CORE PURE / COMPLEX NUMBERS

Loci and Regions in the Argand Diagram

Questions by topic 10 questions · 87 marks

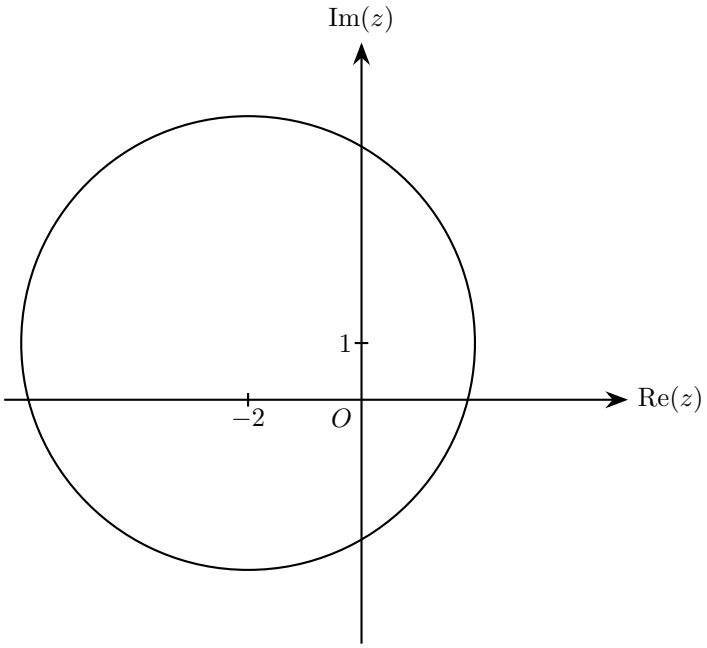
EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Instructions

Cambridge International: this topic belongs to A Level Mathematics (9709), Paper 3 and is assumed knowledge for Further Mathematics 9231.

Questions

Question 1	*6 marks*	2
Question 2	*6 marks*	3
Question 3	*10 marks*	4
Question 4	*11 marks*	5
Question 5	*11 marks*	6
Question 6	*12 marks*	7
Question 7	*7 marks*	8
Question 8	*8 marks*	9
Question 9	*9 marks*	10
Question 10	*7 marks*	11



1. The Argand diagram shows a circle of radius 4. The centre of the circle is the point which represents the complex number $-2+i$.

(a) Use set notation to define the locus of complex numbers, z, represented by points which lie on the circle. **[2]**

The locus L is defined by

$$L = \left\{ z \in \mathbb{C} \;\middle|\; |z-(2+i)| = |z+1-3i| \right\}$$

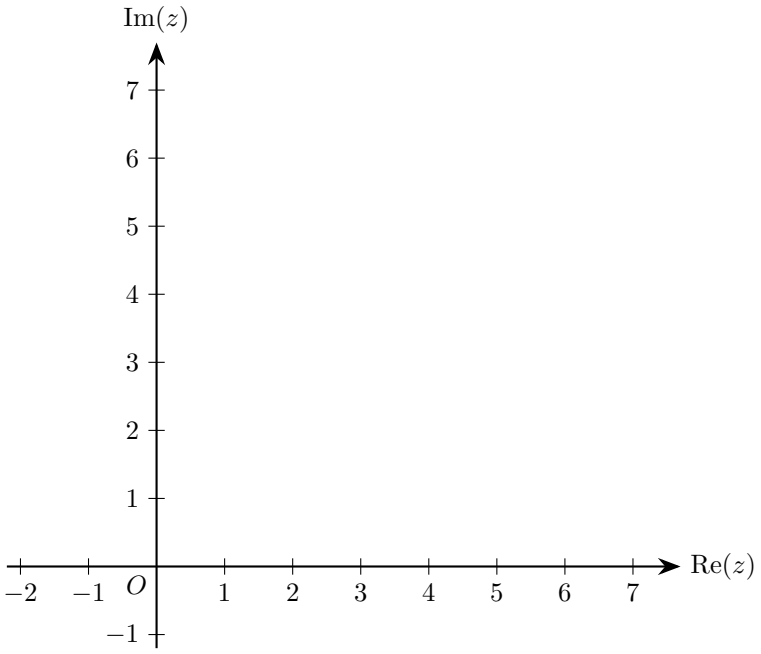
(b) On a copy of the Argand diagram, sketch and label the locus L. **[2]**

You are given that the locus

$$\left\{ z \in \mathbb{C} \;\middle|\; \arg(z+1-2i) = \frac{2\pi}{3},\ \mathrm{Im}(z) = 5 \right\}$$

contains only one number.

(c) Find this number. **[2]**



2. **(a)** Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (2 + 4i)| = 3$$

[2]

(b) Sketch, also on the Argand diagram, the locus of points satisfying the equation

$$\arg(z - 1) = \frac{\pi}{4}$$

[1]

(c) For the complex number w, find the maximum value of $|w|$ such that

$$|w - (2 + 4i)| \leq 3 \qquad \text{and} \qquad 0 \leq \arg(w - 1) \leq \frac{\pi}{4}$$

[3]

3. On an Argand diagram, the locus of points satisfying $|z - 4 + 3i| = 8$ is a circle.

(a) Write down

(i) the coordinates of the centre of this circle **[1]**

(ii) the radius of this circle **[1]**

(b) On an Argand diagram, shade the set of points defined by

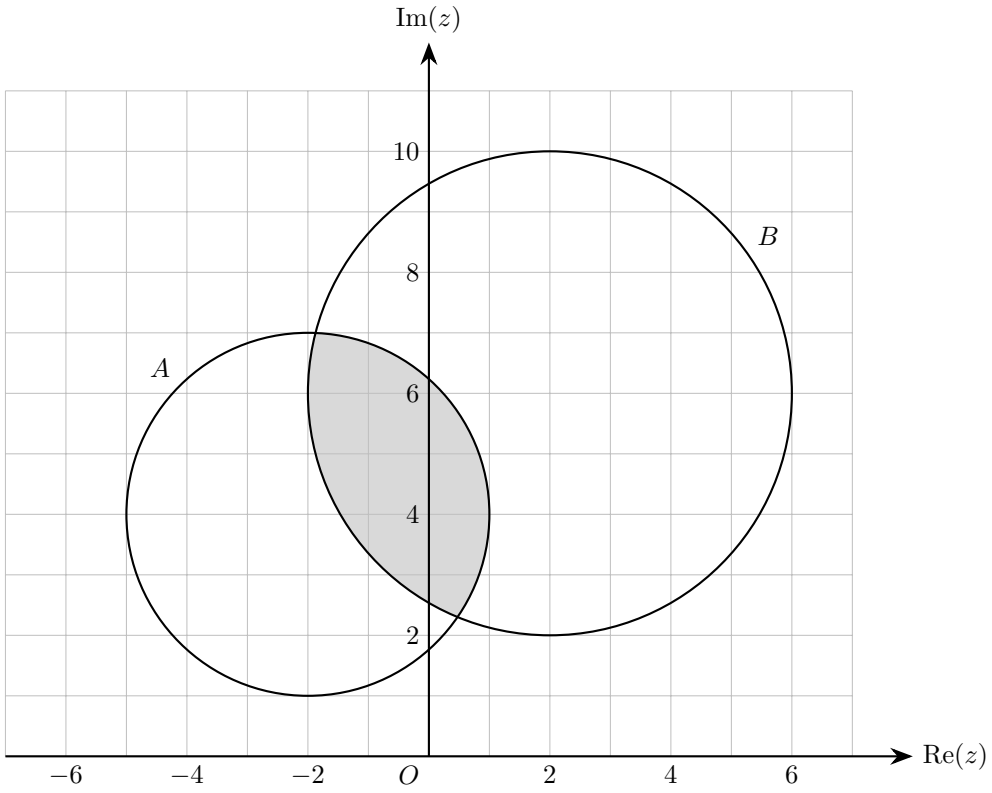
$$|z - 4 + 3i| \leq 8$$ **[1]**

(c) For the set of points defined in part (b), determine the maximum value of $|z|$. **[3]**

The set of points A is defined by

$$A = \left\{ z \in \mathbb{C} \;\middle|\; -\frac{\pi}{4} \leq \arg(z - 3 - 6i) \leq \frac{3\pi}{4} \right\} \bigcap \left\{ z \in \mathbb{C} \;\middle|\; |z - 4 + 3i| \leq 8 \right\}$$

(d) Determine the area of the region defined by A, giving your answer to 3 significant figures. **[4]**



4. **(a)** The diagram above shows an Argand diagram.

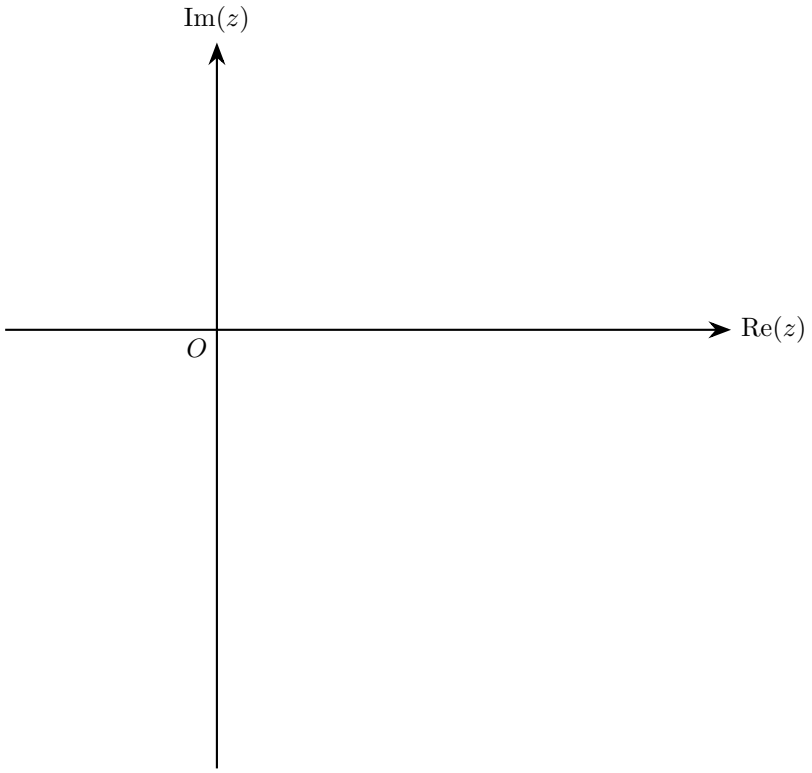
(i) Write down the equation of the locus represented by the circumference of circle A. **[2]**

(ii) Write down the two inequalities that define the shaded region, not including either circumference. **[3]**

(b) (i) Draw an Argand diagram to show the region where

$$-\frac{\pi}{6} < \arg(z + 2 - i) < \frac{\pi}{4}$$ **[3]**

(ii) Determine whether the point $1 + 5i$ lies within this region. **[3]**



5. **(a)** Find the modulus of the complex number $2 + 2\sqrt{3}i$, giving your answer as an integer. **[2]**

(b) The locus of points, L, satisfies the equation $|z - (2 - 2\sqrt{3}i)| = 2$.

(i) Sketch the locus L on the Argand diagram above. **[2]**

(ii) The complex number w lies on L so that $-\pi < \arg w \leq \pi$. Find the greatest possible value of $\arg w$, giving your answer in terms of π. **[2]**

(c) Solve the equation $z^4 = 2+2\sqrt{3}i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[5]**

6. The point P represents a complex number z on an Argand diagram such that

$$|z-(1+i)| = \frac{1}{2}|z-(4+i)|$$

(a) Show that, as z varies, the locus of P is a circle, and give the coordinates of the centre and the radius of the circle. **[5]**

The point Q represents a complex number z on an Argand diagram such that

$$|z-(2+3i)| = |z+2-i|$$

(b) Sketch, on the same Argand diagram, the locus of P and the locus of Q as z varies. **[5]**

(c) On your diagram shade the region which satisfies

$$|z-(1+i)| \leq \frac{1}{2}|z-(4+i)| \quad \text{and} \quad |z-(2+3i)| \geq |z+2-i|$$

[2]

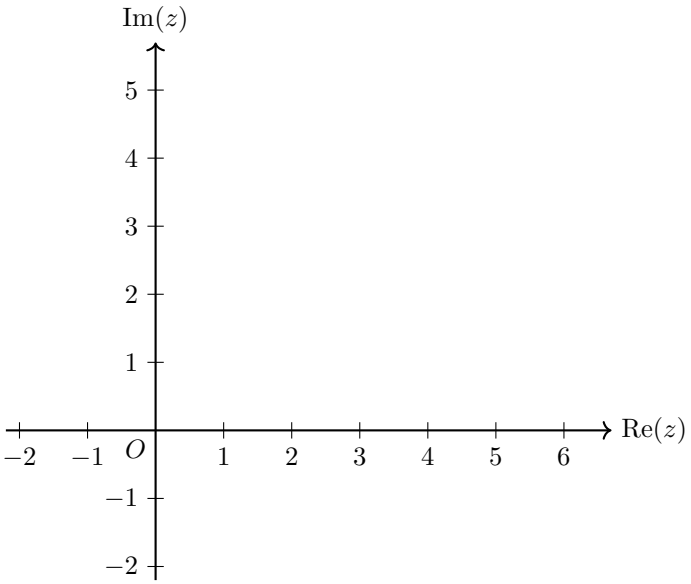
7. On an Argand diagram, the point P representing the complex number z lies on the locus

$$|z - (2 - i)| = 3$$

The point Q representing the complex number w lies on the locus

$$|w + 1 - 11i| = 3|w + 1 - 3i|$$

Find the exact coordinates of the points common to these two loci. **[7]**



8. **(a)** Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (3 + i)| = 1$$ **[2]**

(b) There is a unique complex number w that satisfies both

$$|w - (3 + i)| = 1 \quad \text{and} \quad \arg(w) = \alpha$$

where α is a constant such that $0 < \alpha < \pi$.

(i) Find the value of α. **[2]**

(ii) Find w and express it in the form $r(\cos\theta + i\sin\theta)$. Give each of r and θ to 2 significant figures. **[4]**

9. In an Argand diagram, the point P representing the complex number w lies on the locus defined by

$$\left\{z \in \mathbb{C} \;\middle|\; \arg(z-2+i) = \frac{\pi}{6}\right\}$$

You are given that $\mathrm{Im}(w) = 3$.

(a) Find w. **[2]**

The point P also lies on the locus defined by $\left\{z \in \mathbb{C} \;\middle|\; |z-2-11i| = k\right\}$, where k is a constant.

(b) Find the complex number represented by the other point of intersection of the loci defined by

$$\left\{z \in \mathbb{C} \;\middle|\; |z-2-11i| = k\right\} \quad \text{and} \quad \left\{z \in \mathbb{C} \;\middle|\; \arg(z-2+i) = \frac{\pi}{6}\right\}$$

[7]

10. The locus of points $z = x + iy$ that satisfy

$$\arg\left(\frac{z-(2+i)}{z-(6+5i)}\right) = \frac{\pi}{4}$$

is an arc of a circle C.

(a) On an Argand diagram sketch the locus of z. **[2]**

(b) Explain why the centre of C lies on the line $y = 7 - x$. **[1]**

(c) Determine the radius of C. **[2]**

(d) Determine the coordinates of the centre of C. **[2]**